E7 Spring 2026

Midterm practice problem set #1

(version 2)

Instructions

- Multiple choice questions with circular bullets have a single correct answer, and you should select only one option.
- Multiple choice questions with square bullets may have more than one correct answer, and you should select all that apply.
- The following package imports apply to all problems.

```python
import math
import cmath
import numpy as np
import pickle
from pathlib import Path
```

Errata

- version 2 includes corrections to questions 5.3, 10.1, 16.1, 17.4. Question 17.5 was eliminated.

Week 2: Scalars, `if` clauses, `while` loops

1) General questions

1. Describe in your own words what each of the following built-in functions do: `int`, `float`, `print`, `input`, `type`, `len`, `range`, `enumerate`, `zip`.
2. Name and describe Python's five built-in scalar data types.
3. Name and describe Python's five built-in collection data types.
4. What do the each of the seven arithmetic operations do? (`+`, `-`, `*`, `/`, `**`, `//`, `%`)

2) Mathematical formulas

2.1) Which of the following lines of code correctly evaluates this mathematical formula?

$$\log\left(\frac{1}{2x}\right) + \sin^2(\pi/4)$$

- ◯ `math.log(1/(2*x)) + math.sin(math.pi/4)**2`
- ◯ `math.log(1/(2*x)) + math.sin(pi/4)**2`
- ◯ `math.log(1/2*x) + math.sin(math.pi/4)**2`
- ◯ `math.log(1/(2*x)) + math.sin**2(math.pi/4)`

2.2) Which of the following lines of code correctly evaluates this mathematical formula?

$$\sqrt{x^3} - e^{2x}$$

- ☐ `math.pow(x**3, 1/2) - math.exp(2*x)`
- ☐ `math.sqrt(x**3) - math.e**(2*x)`
- ☐ `math.sqrt(math.pow(x, 3)) - math.exp(2*x)`

2.3) Which of the following lines of code correctly evaluates this mathematical formula? ($j = \sqrt{-1}$)

$$(2 + 3j) + e^{\pi j}$$

- ☐ `2 + 3*j + cmath.exp(math.pi * j)`
- ☐ `2 + 3*j + cmath.exp(math.pi * 1j)`
- ☐ `2 + 3i + cmath.exp(math.pi * 1i)`
- ☐ `2 + 3j + cmath.exp(math.pi * 1j)`

2.4) Which of the following lines of code correctly evaluates this mathematical formula?

$$\frac{a + b(-c)}{d^2 + 1}$$

- ☐ `(a+b*-c)/d**2+1`
- ☐ `(a+b*(-c))/(d**2+1)`
- ☐ `a+b*(-c)/d**2+1`
- ☐ `(a+(b-c))/(d**2+1)`

3) Type conversion

3.1) What is the result of each of these type conversions?

1. `int(math.pi)`
2. `str(250.2)`
3. `float(True)`
4. `int("4.7")`

4) Compound boolean expressions

4.1) Reduce each of the following compound expressions to its boolean value. Show each step, clearly indicating what was evaluated at each stage.

1. `not 2*3/4**4 < 1 or 4%2**3+1 == 4`
2. `5%4**2*(1-2%2)/-4+5`
3. `(3-2*5)**2 - ((6*3)==18) * (4*5-16**1/2)`

5) Tracking code

5.1) What is the value of `x` after executing this code?

```
x = 2
x += x
x *= x
```

5.2) What is the value of `x` after executing this code?

```
x = 0
while x < 10:
    x+=2
```

5.3) What is the value of `x` after executing this code?

```
x=17
if x%2 != 0:
    x += 1
else:
    x -= 1
```

5.4) What is the value of `A` after executing this code?

```
A = 20
B = 40
while B > 10:
    if A % 2 == 0 and B % 2 == 0:
        A /= 2
        B -= 5
    elif A % 2 == 0 or B % 2 == 0:
        A -= 5
        B /= 2
    else:
        B /= 4
```

5.5) What is the value of `sum_xy` after executing this code?

```
x = 1
lim = 3
sum_xy = 0
while x <= lim:
    y = x
    while y <= lim:
        sum_xy += x*y
        y += 1
    x += 1
```

5.6) What is the value of `num` after executing this code?

```
num = 1
while True:
    if num % 5 == 1 and num % 2 == 0:
        break
    num += 1
```

5.7) What is the value of `counts` after executing this code?

```
num = 0
counts = 0
while num < 9:
    num += 1
    if num % 2 == 1:
        counts += 1
        continue
    if num % 3 == 0:
        counts += 1
```

Week 3: Collections, for loops

6) General questions

1. List the sequential collection types.
2. List the immutable collection types.
3. Describe each collection type in your own words.

7) Type conversion

7.1) What is the result of this type conversion?

```
set([1, 2, (1, 2), 1.0, 2, 'a', 'aa', "a"])
```

8) Tracking code

8.1) What is the value of `x` after executing this code?

```
x = 0
for i in range(1,8,2):
    x = x+2
```

8.2) What is the value of `x` after executing this code?

```
A = [1, 2, 3, 4]
i = len(A)
x = []
while i > -len(A):
    i -= 1
    x.append(A[i])
```

8.3) Which statements will be true after running the following code?

```
s = "mississippi"
prev = {}
ans = []
for i, x in enumerate(s):
    if x in prev.keys():
        ans.append(prev[x])
    else:
        ans.append(-1)
    prev[x] = i
```

- ☐ `ans` will be a list containing strings and integers.
- ☐ `prev` will be a dictionary whose keys are strings and values are integers.
- ☐ `prev` will have length 4.

8.4) What is the value of `lst` after executing this code?

```
arr = ['a', 'b', 'c']
lst = [2, 1, 0]
for i, x in enumerate(range(len(arr))):
    lst[i] = x
```

8.5) What is the value of `s` after executing this code?

```
arr = [1, 2, 3]
s = 0
for i, x in enumerate(arr):
    s += x ** i
```

8.6) What is the value of `s` after executing this code?

```
A = [1, 6, 20]
B = [2, 3, 5]
s = 0
for x, y in zip(A, B):
    s += x / y
```

9) Indexing

9.1) Given

```
A = [[5,2,9,3,0],[-1,4,-7,1,8],[6,-2,-9,7,-3]]
```

What is the result of the following selections?

1. `A[1,::2]`
2. `A[::-1,::-1]`
3. `A[[0,2,1],[1,2,1]]`

9.2) Continuing with the array `A` from the previous question, write expressions to select each of the following values from `A`.

1. `0`
2. `[4,-7,1]`
3. `[-3,-9,-6]`

9.3) Choose the correct way to access the 30 from the following tuple

```
atuple = ("red", [10, 30, 50], (2, 4, 6))
```

- ◯ `atuple[1:2](1)`
- ◯ `atuple[1:2][1]`
- ◯ `atuple[1:2][2]`
- ◯ `atuple[1][1]`

10) Functions

10.1) Given this function, what is the result of each of the calls listed below?

```
def my_function(a,b=1,c=0):
        return (a+c)/b
```

1. `my_function(1,1,2)`
2. `my_function(b=1,a=1,3)`
3. `my_function(b=1,a=1,c=1)`
4. `my_function(4,c=2,b=2)`

11) Dictionaries

11.1) Which of the following queries returns Priya Sharma's grade?

```
grades_by_id = {

        "S9381" : {
                "last_name" : "Martinez",
                "first_name": "Olivia",
                "grade": 41
        },

        "S2481" : {
                "last_name" : "Reynolds",
                "first_name": "Ethan",
                "grade": 36
        },

        "S0683" : {
                "last_name" : "Sharma",
                "first_name": "Priya",
                "grade": 37
        }
}
```

- ◯ `grades_by_id["Sharma"]["grade"]`
- ◯ `grades_by_id[Sharma][grade]`
- ◯ `grades_by_id["S0683"]["grade"]`
- ◯ `grades_by_id["grade"]["S0683"]`
- ◯ `grades_by_id[37]`

12) Comprehensions

12.1) Given

```
A = [1, 2, 3, 4, 5]
```

use a comprehension to construct the list of the squares of values in `A`.

12.2) Given

```
A =  [1, 2, 3, 4, 5, 6, 7, 8]
```

use a comprehension to construct the list of the negatives of the even values in `A`.

12.3) Given

```
names = ["Gertrude", "Gaston", "Pedro", "Lisa", "Priscilla"]
```

use a comprehension to find the unique set of the first letters of names in `A`, in lower case.

12.4) Given

```
A = ["elephant","galaxy","mountain","tree","sun"]
```

use a comprehension to find the unique lengths of the words in `A` that have more than five letters.

Week 4: NumPy

13) General questions

1. List some of the important differences between NumPy arrays and Python lists.
2. List some of the pros and cons of NumPy arrays with respect to Python lists.
3. In which situation would you use one or the other?
4. Why do we typically use methods such as `np.zeros` or `np.empty` to initialize NumPy arrays, instead of growing them from an empty one as we do with lists?
5. When is it a good idea to initialize a NumPy array with `np.empty`?
6. What is the main difference between Python's `range` and NumPy's `np.arange`?

14) Array aggregations

14.1) Use NumPy aggregations to evaluate this formula. Assume that `x` is a 1D NumPy array with n elements.

$$\sum_{i=1}^{n} |x_i|$$

14.2) What is the *shape* of `D` after executing this code?

```
A = np.ones((3, 360))
D = np.mean(A, axis=0)
```

14.3) Suppose you have a 2D NumPy array `M` filled with boolean values. Use NumPy aggregations to find each of the following with a single line of code.

1. Whether there are any `True` values in `M`.
2. Whether there are any `False` values in `M`.
3. Whether each column of `M` has any `True` values.
4. Whether each row of `M` consists of only `False` values.
5. The number of `True` values in `M`.
6. The number of `True` values in each column of `M`.

15) Operations and broadcasting

15.1) What is the value of `A` after executing this code?

```
A = np.array([[0,7,8,9]]) % 7
```

15.2) What is the value of `A` after executing this code?

```
A = np.array([[1, 2], [3, 4]]) * np.array([1, 2])
```

15.3) What is the value of `A` after executing this code?

```
A = np.ones((2, 3)) + np.array([1, 2, 3])
```

15.4) What is the value of `B` after executing this code?

```
A = np.array([1, 2, 3]) + np.array([[1], [2], [3]])
B = np.any(np.sum(A, axis=1) == 15)
```

16) Indexing arrays with integers, slices, and lists

16.1) Given

```
B = np.array([ 5,  2,  9,  3,  0])
```

for each of the arrays listed below, write two distinct ways that they could be obtained using an expression of the form `B[sel]` (where `sel` is any valid selector).

1. `0`
2. `np.array([0,3,9])`
3. `np.array([5, 9, 0])`
4. `np.array([7])`
5. `np.array([5, 2, 9])`
6. ```
   np.array([[ 5,  2,  9,  3,  0],
             [-1,  4, -7,  1,  8]])
   ```
7. ```
   np.array([[-2],
             [ 4],
             [ 2]])
   ```

16.2) Given

```
A = np.array([[ 5,  2,  9,  3,  0],
              [-1,  4, -7,  1,  8],
              [ 6, -2, -9,  7, -3]])
```

write expressions of the form `A[sel1,sel2]` that return each of the following values.

1. `7`
2. `np.array([7])`
3. `np.array([5, 2, 9])`
4. ```
   np.array([[ 5,  2,  9,  3,  0],
             [-1,  4, -7,  1,  8]])
   ```
5. ```
   np.array([[-2],
             [ 4],
             [ 2]])
   ```

16.3) Continuing with the array `A` from the previous question, what is the result of the following selections?

1. `A[1,::2]`
2. `A[::-1,::-1]`
3. `A[[0,2,1],[1,2,1]]`

16.4) What is the value of `d` after executing this code?

```
a = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
b = a[::3]
c = a[2::2]
d = b+c
```

16.5) Which of the following options select the value

```
array([[ 6,  8],
       [14, 16]])
```

from the 2D array `M` define below?

```
M = np.array([[1,   2,  3,  4],
              [5,   6,  7,  8],
              [9,  10, 11, 12],
              [13, 14, 15, 16]])
```

- ◯ `M[(1,3),(1,3)]`
- ◯ `M[1:2,1:2]`
- ◯ `M[1::2,1::2]`
- ◯ `M[(1::2,1::2)]`

17) Indexing arrays with boolean masks

17.1) What is the value of `Z` after executing this code?

```
L = np.linspace(0,10,11)
T = L[L%3==0]
Z = T[-1]**T[1]
```

17.2) What is the value of `scores` after executing this code?

```
scores = np.array([25, 36, 49, 64, 81, 100])
mask = scores < 80
scores[mask] = 10 * np.sqrt(scores[mask])
```

17.3) What is the value of `d` after executing this code?

```
a = np.arange(10) ** 2
d = np.any( a[ (a <= 80) & (a >= 65) ] )
```

17.4) What is the value of `B` after executing this code?

```
A = np.array([ [1, 2, 3],
               [4, 5, 6]])
B = np.all( A[:, A[1, :] >= 6] < 4 )
```

18) Creating arrays

18.1) Which of the listed options produce the following NumPy array?

```
np.array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. ])
```

- ☐ `np.linspace(0,1,11)`
- ☐ `np.arange(0,1.1,0.1)`
- ☐ `np.array([0.1*i for i in range(11)])`
- ☐ `np.array(range(11))/10`